

SPEED: Scalable, Precise, and Efficient Concept Erasure for Diffusion Models — An Empirical Analysis

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Repository: github.com/Vedant2100/concept-erasure-analysis

Abstract

Large-scale text-to-image diffusion models learn to generate harmful, copyrighted, or otherwise undesirable content from internet-scale training data. *Concept erasure* surgically removes specific visual concepts from a pretrained model's weights without retraining from scratch and without post-hoc filters that can be bypassed. This report traces the research lineage from Erasing Concepts from Diffusion Models (ESD; Gandikota et al., ICCV 2023) through to SPEED (Li et al., ICLR 2026), identifies SPEED as the current open-source frontier for training-free concept erasure, and presents a battery of targeted empirical probes that expose two concrete, reproducible limitations: (1) **rank saturation collapse** under concentrated mass erasure of semantically entangled concepts, and (2) **lexical overfitting** that allows trivial paraphrase-based evasion of style erasure. We additionally present an ablation study demonstrating that SPEED's own Prior Knowledge Refinement (DPA) module actively accelerates the rank saturation failure, and a methodological audit documenting three measurement pitfalls that would have produced false-positive findings.

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1. Introduction

The proliferation of open-source text-to-image diffusion models — Stable Diffusion (Rombach et al., 2022), SDXL (Podell et al., 2023), and their derivatives — has created an urgent need for post-hoc mechanisms to prevent the generation of specific visual concepts. Applications include removing copyrighted artistic styles, preventing the generation of recognizable likenesses without consent, and suppressing the production of harmful or illegal content.

The concept erasure literature has evolved rapidly from inference-time steering methods (Safe Latent Diffusion; Schramowski et al., 2023) to direct weight editing (ESD; Gandikota et al., 2023), closed-form projection (UCE; Gandikota et al., 2024), and training-free null-space methods (SPEED; Li et al., 2026). Each generation of methods addresses shortcomings of its predecessors, but introduces new failure modes.

This project fulfills the EE 243 assignment requirements:

1. **Trace the lineage.** We map the intellectual genealogy from the foundational Stable Diffusion architecture through to SPEED, identifying the key technical transitions.
2. **Find the frontier.** We identify SPEED (ICLR 2026) as the current state-of-the-art open-source method for efficient concept erasure.
3. **Articulate strengths.** We describe SPEED's genuine contributions: training-free closed-form editing, null-space preservation guarantees, and extreme computational efficiency.
4. **Expose limitations through experiments.** We design and execute four targeted empirical probes that locate two concrete, reproducible bottlenecks in SPEED's architecture.

No models were trained or fine-tuned for this analysis. All experiments use the publicly released SPEED and ESD checkpoints and the CompVis/stable-diffusion-v1-4 base model.

2. Research Lineage and the Frontier

ESD was not the first attempt to make generative models safer, nor the last. The field converged on weight editing after finding dataset filtering too expensive and inference-time steering inadequate. The lineage from foundations to the current frontier proceeds as follows.

2.1 Foundational Architecture

Stable Diffusion (Rombach et al., CVPR 2022) introduced the latent diffusion architecture that all subsequent concept erasure methods target. In this architecture, the text prompt conditions image generation through cross-attention layers in the U-Net denoiser, where token embeddings are projected through Key and Value matrices (W_K , W_V) that tell each spatial region what the text is asking for.

2.2 First-Generation Erasure: Inference-Time Steering

Safe Latent Diffusion (SLD) (Schramowski et al., CVPR 2023) modified the classifier-free guidance equation at inference time to steer the denoising trajectory away from unsafe latent subspaces. This approach is practical because it requires no retraining, but adds computational overhead during inference and is trivially bypassed if users have raw parameter access.

2.3 Second-Generation Erasure: Weight Editing

Erasing Concepts from Diffusion Models (ESD) (Gandikota et al., ICCV 2023) was the first practical weight-editing method. ESD fine-tunes the U-Net's cross-attention parameters using inverted classifier-free guidance. It comes in two variants:

- **ESD-x**: Fine-tunes only the cross-attention Key/Value matrices, providing more targeted erasure.
- **ESD-u**: Fine-tunes all unconditional layers, providing broader but less precise erasure with greater collateral damage.

Concept Ablation (CA) (Kumari et al., ICCV 2023) concurrently proposed learning to shift the target concept's distribution to match a safe "anchor" concept by minimizing KL divergence.

2.4 Third-Generation Erasure: Closed-Form Projection

Unified Concept Editing (UCE) (Gandikota et al., WACV 2024) replaced iterative gradient descent with a closed-form system of linear equations, directly projecting erased concepts to neutral targets while preserving protected concepts. This eliminated the need for iterative fine-tuning but introduced scalability limits.

MACE (Lu et al., CVPR 2024) and **RECE** (Gong et al., ECCV 2024) extended UCE's approach to handle multiple concepts simultaneously, using LoRA adapters and iterative closed-form updates respectively.

2.5 The Current Frontier

SPEED (Li et al., ICLR 2026) represents the current state-of-the-art along the efficiency and closed-form branch. It computes weight updates that redirect target concept embeddings toward neutral ones, then projects those updates onto the null-space of a large retain set (~1,700 artists), achieving training-free editing in approximately 5 seconds.

2.6 Parallel Branches

From ESD, the literature diverges into several distinct branches that we do not evaluate in detail but acknowledge for completeness:

Branch	Representative Papers	Focus
Efficiency & Closed-Form	UCE → RECE → SPEED	Training-free, fast editing
Mass Erasure	MACE → DyME → ETC (Seo et al., CVPR 2026)	Scaling to hundreds/thousands of concepts
Robustness	AdvUnlearn (Zhang et al., NeurIPS 2024), RACE	Adversarial training against prompt attacks

Branch	Representative Papers	Focus
Localization	GLoCE (Lee et al., CVPR 2025), LACE	Spatial precision, avoiding global collateral damage

3. Target Paper: SPEED

3.1 Mechanism

SPEED operates on the cross-attention Key and Value projection matrices (W_K , W_V) in the U-Net. For a target concept c_{target} to be erased and a neutral concept c_{neutral} :

- Embedding Computation.** Compute the CLIP text embeddings e_{target} and e_{neutral} for the target and neutral concepts.
- Weight Update.** Compute a weight update ΔW that redirects the target embedding toward the neutral one: $\Delta W = (e_{\text{neutral}} - e_{\text{target}}) \otimes e_{\text{target}}^T$.
- Null-Space Projection.** Project ΔW onto the null-space of the retain set's activation matrix C_0 . If $C_0 = [c_1, c_2, \dots, c_R]$ is the matrix of retain concept embeddings, then the null-space projector is $P_{\text{null}} = I - C_0 C_0^T (C_0 C_0^T)^{-1} C_0$, and the final update is $\Delta W_{\text{final}} = \Delta W \cdot P_{\text{null}}$.
- Prior Knowledge Refinement (DPA).** SPEED augments the retain set with perturbed "fake" embeddings to increase the coverage of the retain space, using a data-free prior augmentation strategy.
- Importance Pruning.** SPEED selectively prunes low-importance weight updates to maintain model utility.

3.2 The Rank Saturation Dilemma

SPEED's authors acknowledge but never empirically demonstrate a fundamental limitation: the null-space of $C_0 C_0^T$ has **finite rank**. As the protected set grows, the available subspace for non-destructive editing shrinks — potentially toward the trivial null-space $\{0\}$. They write:

"As R increases, $C_0 C_0^T$ gradually reaches full rank, its null space narrows and reduces to the trivial null space $\{0\}$."

Our Experiment 3 locates precisely where this begins to bite.

4. Algorithmic Strengths

SPEED improves upon MACE and UCE significantly along three axes:

Extreme Efficiency. SPEED relies on training-free null-space editing. It iteratively projects out target concept directions via importance pruning and embedding constraints, scaling to hundreds of concepts in approximately 5 seconds on a single GPU. By comparison, ESD requires iterative fine-

tuning (~10 minutes per concept), and MACE requires training individual LoRA adapters.

Precision. By operating strictly in the null-space of the retain set's activation matrix, SPEED systematically minimizes collateral damage to non-target concepts. Earlier methods like ESD-u suffer from significant global degradation because they modify unconditional layers that affect all concepts indiscriminately.

Open-Source Accessibility. Unlike many concurrent mass-erasure papers, SPEED provides publicly released checkpoints and a clean codebase, enabling immediate empirical evaluation without requiring expensive compute for fine-tuning. This is what made our analysis possible.

4.1 Strengths Demonstrated

In our own experiments, SPEED's strengths are confirmed:

- In the sparse multi-concept erasure regime ($N=3$), canary artists drift no more than stylistically unrelated controls (Section 6), confirming the null-space guarantee holds under light load.
 - Perceptual fidelity (LPIPS) of protected neighbors is strictly bounded below the drift of style-far controls (Section 10), confirming SPEED preserves not just semantic labels but pixel-level structural quality.
 - Erasure fires reliably: erased artists show CLIP drift of 0.332–0.388, confirming genuine concept removal (Sections 6–7).
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5. Experimental Methodology

5.1 Infrastructure

All experiments were executed on GPU cluster nodes via SLURM. The base model is CompVis/stable-diffusion-v1-4. SPEED checkpoints were obtained from the official repository. ESD-x checkpoints were obtained from the ESD project's public release.

5.2 Generation Protocol

- **Precision:** All generation runs in `fp32` to prevent VAE NaN artifacts that occur in `fp16`.
- **Safety Checker:** Disabled (`safety_checker=None`). Stable Diffusion's NSFW safety checker blanks flagged generations to solid black, which poisons all distance metrics. Disabling it does not change seeds, so paired baseline-vs-edited comparisons remain valid.
- **Black Frame Guard:** Every generated image is checked for near-black content (mean pixel intensity < 10). Black frames are excluded from all analyses, and the count of excluded frames is reported alongside every metric.
- **Seeding:** Each prompt is generated at 4 deterministic seeds (0, 1, 2, 3) with 50 DDIM inference steps and guidance scale 7.5. All metrics are averaged over valid seeds.

5.3 Metrics

CLIP Image-to-Image Drift

The primary metric throughout this report is CLIP image-to-image drift:

```
drift = 1 - cos(CLIP_image(baseline), CLIP_image(edited))
```

where baseline and edited are the same prompt at the same seed, generated by the unmodified and edited models respectively. CLIP responds to style rather than pixel layout, making it the appropriate metric for measuring stylistic change.

Why not pixel MSE? Pixel-space MSE is dominated by ordinary seed-to-seed composition variation. In early experiments, a *retained, protected* artist (Cézanne) scored $MSE > 10,000$ on some seeds while a supposedly "damaged" neighbor scored ~ 440 . MSE cannot distinguish style damage from diffusion stochasticity.

CLIP Image-to-Text Similarity ("Van-Gogh-ness")

For the paraphrase evasion experiment (Section 11), we compute:

```
VG-ness = cos(CLIP_image(generated), CLIP_text("a painting in the style of Van Gogh"))
```

This tracks whether a generated image carries Van Gogh's stylistic signature regardless of the prompt used to generate it. Style-far controls (ukiyo-e, Rembrandt, Hokusai) sit at approximately 0.18, establishing the "not Van Gogh" floor.

LPIPS (Learned Perceptual Image Patch Similarity)

For the fidelity robustness check (Section 10), we compute LPIPS between baseline and edited images to detect perceptual degradation that semantic metrics like CLIP might miss.

5.4 Experimental Controls

Every experiment includes two categories of controls:

Control Type	Artists	Purpose
Canary (style-adjacent)	Gauguin, Seurat, Pissarro	Retained impressionists held out of every erasure set. If the null-space leaks, these should show elevated drift.
Style-far control	Rembrandt, Hokusai	Retained but stylistically distant from the erased cluster. Calibrates the noise floor: how much drift is simply "a bigger edit perturbs everything a little."

A canary's drift is interpretable only relative to the style-far controls. If a canary drifts no more than the controls, the null-space is holding. If it drifts significantly above them, the leak is real.

6. Experiment 1: Sparse Multi-Concept Erasure (Positive Control)

6.1 Design

We erased three stylistically diverse painters simultaneously — **Van Gogh, Picasso, and Monet** — and measured drift on three held-out impressionist canaries (Gauguin, Seurat, Pissarro) and two style-far controls (Rembrandt, Hokusai). If the null-space leaked under multi-concept pressure, the canaries would show elevated drift relative to the controls.

6.2 Results

Concept	Role	CLIP Drift
Gauguin	Neighbor (canary)	0.109
Seurat	Neighbor (canary)	0.049
Pissarro	Neighbor (canary)	0.076
Rembrandt	Control (style-far)	0.114
Hokusai	Control (style-far)	0.063

6.3 Analysis

The canaries drift *no more than* the style-far controls. Gauguin (0.109) is statistically indistinguishable from Rembrandt (0.114), a painter with nothing in common with the targets. There is no concentrated leakage onto the neighbors. For sparse, mixed erasure, SPEED's null-space does exactly what it advertises.

This is an honest negative result — and it told us the limitation, if one existed, had to live somewhere harder. The null-space guarantee holds when the erased concepts are stylistically diverse and few in number.

7. Limitation 1: Rank Saturation Collapse Under Concentrated Mass Erasure

7.1 Motivation

SPEED's own evaluation erases either many *diverse* concepts (100 different celebrities) or single painters in isolation. It never erases many artists of a single artistic movement and measures a held-out artist of that same movement. This is precisely the regime where rank saturation — the fundamental capacity limit SPEED's authors acknowledge but never test — should begin to bite.

7.2 Design

We erased a growing, tightly-correlated cluster of **impressionists** — $N = 5, 10, 20$, and 40 — while strictly holding three canaries (Gauguin, Seurat, Pissarro) out of every erasure set. Each list is nested inside the next, so the only variable is how many same-movement concepts we pile on. The

impressionist lists were constructed from art-historical membership lists to ensure dense semantic entanglement.

7.3 Results

CLIP drift vs. baseline, 4 seeds per cell, zero corrupt frames:

Concept	Role	N=5	N=10	N=20	N=40
Renoir	Erased (sanity)	0.354	0.354	0.361	0.332
Pissarro	Canary — core impressionist	0.052	0.126	0.165	0.253
Gauguin	Canary — post-impressionist	0.083	0.119	0.110	0.128
Seurat	Canary — pointillist	0.044	0.120	0.076	0.131
"impressionist oil painting"	Supertype capability	0.035	0.070	0.049	0.036
Rembrandt	Control — style-far	0.050	0.094	0.132	0.113
Hokusai	Control — style-far	0.036	0.040	0.074	0.081

7.4 Analysis

One row breaks away from the pack. **Pissarro** climbs monotonically — 0.052 → 0.126 → 0.165 → **0.253** — until, at N=40, he has drifted to roughly *double* the style-far controls. The degradation is visible: his soft, broken-brushwork impressionism flattens into bolder, more saturated, less characteristic forms. Gauguin, held out alongside the same cluster, stays put.

Key observations:

1. **Monotonic escalation.** Pissarro's drift increases with every addition to the erasure set. This rules out random noise (which would fluctuate) and points to a systematic, load-dependent failure.
2. **Selectivity.** Only the most entangled canary fails. Gauguin (post-impressionist, cloisonnist planes) and Seurat (pointillist dots) remain at control levels. The failure is not a global collapse but a targeted leak at the single weakest point.
3. **Supertype preservation.** The broad "impressionist oil painting" capability is untouched (0.036 at N=40). SPEED does not destroy the artistic movement — it springs a leak at its single most entangled member.
4. **Erasure still fires.** Renoir (erased at all N values) maintains high drift (0.332–0.361), confirming the erasure mechanism itself does not degrade.

Finding: Under concentrated mass erasure of a single dense semantic cluster, SPEED's null-space protection fails for the single retained concept most entangled with what was erased, while

stylistically distinct neighbors and the broader style capability survive. This is a precise, locatable limitation — not a catastrophic failure.

8. Geometric Analysis: Why Only One Neighbor Failed

The failure is geometric, not random. SPEED pushes its edit along the *shared direction* of everything in the erased set, while keeping that edit orthogonal to retained concepts. When you erase 40 impressionists, that shared direction collapses onto one thing — soft, plein-air impressionist landscape — because that is what most of those 40 painters share.

Place each held-out neighbor relative to that direction:

- **Pissarro** is that direction: the most prototypical impressionist of the three, near-indistinguishable from the Monets and Sisleys being erased. When the null-space runs low on degrees of freedom, he is the one concept it cannot keep orthogonal.
- **Gauguin** (flat cloisonnist planes) sits *off* that axis; his distinctive technique gives the projection room to protect him.
- **Seurat** (rigid pointillist dots) sits *off* that axis for the same reason.

This is also why only Pissarro's drift *grows with N* — every additional impressionist reinforces the exact direction he lives on.

Caveat: This is a single collapsing artist, so the geometric story is the best-supported explanation rather than a proven one. The clean confirmation would be to invert the experiment — erase 40 *pointillists* and predict that Seurat becomes the casualty while Pissarro survives.

9. Ablation: The Refinement Contradiction

9.1 Hypothesis

If rank saturation caused Pissarro's collapse at $N=40$, it exposes a significant internal contradiction in SPEED's architecture. SPEED uses **Prior Knowledge Refinement (DPA)**, which generates perturbed, "fake" retain concept embeddings to artificially increase the coverage of the retain space. But adding vectors to the retain set *consumes the rank budget* and shrinks the null-space.

We hypothesized that DPA, designed to protect the prior, was actually accelerating the collapse under load.

9.2 Design

We repeated the $N=40$ impressionist erasure under three configurations:

1. **Full Method (DPA On):** Standard SPEED with all refinement enabled.
2. **DPA Off:** Disabled the data-free prior augmentation (`aug_num=0`), retaining only the original retain set.
3. **Zero Refinement:** Disabled all refinement machinery entirely.

9.3 Results

Concept	Role	Full Method (DPA On)	DPA Off	Zero Refinement
Pissarro	Canary (Leaker)	0.253	0.167	0.141
Gauguin	Canary	0.128	0.109	0.102
Rembrandt	Control	0.113	0.187	0.084

9.4 Analysis

By turning *off* the safety feature, Pissarro's drift plummeted from a devastating 0.253 back down to 0.141 (barely above the noise floor). DPA artificially inflates the retain rank, starving the null-space of the exact degrees of freedom it desperately needs under dense entanglement. SPEED's preservation machinery actively accelerates its own capacity collapse.

An honest caveat on the noise floor: The ablation is not perfectly smooth. In the intermediate "DPA Off" column, the style-far control (Rembrandt) spikes to 0.187, eclipsing Pissarro (0.167). This implies that turning off DPA initially destabilizes the model globally, introducing systemic noise rather than just cleanly fixing the leak. However, turning off the refinement machinery entirely ("Zero Refinement") eliminates this systemic noise, stabilizing the controls back down to 0.084 while locking in the repair to Pissarro. The clean conclusion holds at the "Zero Refinement" endpoint, but the intermediate state is noisier than a clean monotonic story would suggest.

Finding: SPEED's DPA module, designed to improve retain-set coverage, actively accelerates rank saturation collapse under concentrated erasure. Disabling it reduces Pissarro's drift by 44% (0.253 → 0.141).

10. Robustness Check: Fidelity vs. Identity

10.1 Motivation

A common vulnerability in concept erasure is creating a "semantic illusion" — fooling the text encoder (CLIP) into recognizing a retained concept while quietly trashing the pixel-level image quality with artifacts, blur, and structural breakage. To test whether SPEED's preservation of neighbors was such an illusion, we re-scanned the multi-concept erasure images using a pixel-level perceptual metric.

10.2 Method

We computed **LPIPS** (Learned Perceptual Image Patch Similarity) between baseline and edited images for all canaries and controls in the 3-concept erasure regime.

10.3 Results and Finding

SPEED completely passed the test. The perceptual damage to the protected canaries was strictly **bounded below** the baseline drift of the style-far controls:

Concept	Role	LPIPS
Pissarro	Canary	0.244
Seurat	Canary	0.260
Rembrandt	Control (style-far)	0.331
Hokusai	Control (style-far)	0.272

SPEED does not just preserve the semantic label of its neighbors — it preserves their true, pixel-level structural fidelity. The null-space guarantee operates at the perceptual level, not just at the CLIP embedding level.

11. Limitation 2: Evasion Vulnerability via Lexical Overfitting

11.1 Motivation

A true safety or concept-erasure mechanism must be robust against evasion. SPEED's paper evaluates robustness against *nudity* prompts using adversarial attacks (MMA, Ring-A-Bell, UnlearnDiff), but crucially leaves its **style erasure** unevaluated against simple paraphrasing.

11.2 Hypothesis

SPEED's null-space projection is mathematically precise but **lexically overfit**. It ties the erasure strictly to the text embeddings of the exact target tokens (e.g., "Van Gogh"). If a user describes the style without using the exact name, the erasure should be trivially bypassed — because the paraphrased prompt's embedding occupies a different direction in text space, one that was never targeted by the projection.

11.3 Design

We tested three prompt tiers, comparing **Baseline SD 1.4**, **SPEED** (Van Gogh erased), and **ESD-x** (Van Gogh erased):

- **Named:** "a painting in the style of Van Gogh" (sanity check: must be suppressed)
- **Named Artwork:** "Starry Night by Vincent van Gogh"
- **Descriptive Paraphrase:** "a painting of golden wheat fields under a turbulent blue sky with heavy expressive brushwork"
- **Scene Paraphrase:** "a painting of a cafe terrace at night with bold complementary yellow and blue colors, thick textured paint"
- **Style Paraphrase:** "a post-impressionist landscape with swirling rhythmic brushstrokes and vivid complementary colors"
- **Abstract Paraphrase:** "a landscape painting where the brushwork conveys raw emotional turmoil, with bold colors and visible thick paint"

Each prompt was generated at 4 seeds with all three methods (Baseline, SPEED, ESD-x). We

measured both Van-Gogh-ness (CLIP image-to-text similarity to "a painting in the style of Van Gogh") and image drift (CLIP image-to-image distance from the unerased baseline).

11.4 Results

Prompt	Tier	VG-ness: Base → SPEED	SPEED Drift	VG-ness: ESD	ESD Drift
"style of Van Gogh"	Named	0.255 → 0.213 (suppressed)	0.347	0.204	0.388
"Starry Night by Van Gogh"	Named	0.278 → 0.269 (leaked)	0.067	0.204	0.289
"golden wheat... brushwork"	Paraphrase	0.243 → 0.244 (leaked)	0.019	0.221	0.117
"cafe terrace complementary ..."	Paraphrase	0.246 → 0.247 (leaked)	0.029	0.226	0.110
"post-impr swirling rhythmic..."	Paraphrase	0.242 → 0.231 (leaked)	0.046	0.199	0.174

11.5 Analysis

The results demonstrate a stark contrast:

1. **Named prompt (sanity check).** Both SPEED and ESD successfully suppress the style. SPEED's Van-Gogh-ness drops from 0.255 to 0.213 (approaching the 0.18 floor), and image drift is high (0.347), confirming the erasure fires.
2. **Paraphrase prompts (the vulnerability).** SPEED's Van-Gogh-ness on every paraphrase stays at or near the baseline level (~ 0.24), and its image drift is near zero (0.019–0.046). The style leaks straight through the erasure, essentially untouched.
3. **ESD-x comparison.** ESD-x reduces paraphrase Van-Gogh-ness substantially further (0.199–0.226) and alters the images far more (drift 0.110–0.174). ESD-x does not achieve complete erasure on paraphrases either (scores remain above the 0.18 floor), but it suppresses the style significantly more than SPEED.

Key insight: SPEED erases the *token*, not the *style*. Because SPEED's null-space projection operates strictly on the text embedding of the exact target string, it has no effect on paraphrased descriptions that evoke the same visual features through different tokens. ESD-x, by contrast, fine-tunes the actual U-Net cross-attention weights, altering the model's internal representation of the style rather than just its response to specific text tokens.

Finding: SPEED's style erasure is trivially bypassed by describing the target style without using the target name. This is a fundamental architectural limitation of token-level null-space projection methods, not a bug.

12. Measurement Pitfalls and Methodological Corrections

Getting to a trustworthy number was harder than it looks. Three measurement traps each produced a convincing-but-wrong result before we caught them — and every safeguard in the pipeline above exists because of one of them.

12.1 Pixel MSE Cannot Tell Damage from Noise

Our first instinct was to measure neighbor damage with pixel-space MSE between baseline and erased images. It failed badly: a *retained, protected* artist (Cézanne) scored MSE above **10,000** on some seeds, while a supposedly "damaged" neighbor scored ~440. MSE is dominated by ordinary seed-to-seed composition variation and cannot distinguish real erasure damage from diffusion stochasticity. This is why every number in this report uses CLIP image-to-image drift, which responds to style rather than pixel layout.

12.2 The NSFW Safety Checker Manufactured a Fake Collapse

An early concentrated-erasure run showed a dramatic "Gauguin collapses to 0.267 drift" signal — entirely an artifact. Stable Diffusion's NSFW safety checker blanks flagged generations to *solid black*, and it fires frequently on nude-heavy painters like Gauguin (his Tahitian series). A black frame compared against a normal painting maxes out CLIP distance, manufacturing a fake collapse. We fixed it at the source: generation now runs in fp32 with the safety checker disabled, and the analyzer explicitly excludes any black frame and reports how many valid seeds each number rests on.

12.3 A Neighbor "Suppression" Claim That Reversed Under Measurement

An initial single-concept framing claimed SPEED visibly suppressed a neighbor's color saturation. Measuring it directly told the opposite story — saturation actually rose slightly. It was confirmation bias reading damage into noise. Discarding that false start is what pushed us toward the controlled, multi-seed, control-calibrated experimental design used throughout this report.

12.4 Why This Matters

Each pitfall would have produced a *more dramatic* headline than the real finding. The selective Pissarro leak survived all three corrections — which is precisely why we trust it. The controls, the seed averaging, the CLIP metric, and the black-frame exclusion are not decoration; they are the difference between a real limitation and an artifact.

13. Discussion

13.1 Summary of Findings

Finding	Status	Evidence
Null-space holds under sparse erasure	Confirmed	Canaries $\leq$ controls at N=3 (Section 6)

Finding	Status	Evidence
Perceptual fidelity preserved	Confirmed	LPIPS canaries < controls (Section 10)
Rank saturation under concentrated erasure	Limitation 1	Pissarro drift monotonically increases to 2× controls at N=40 (Section 7)
DPA accelerates rank saturation	Confirmed	Disabling DPA reduces leaker drift by 44% (Section 9)
Lexical overfitting enables paraphrase evasion	Limitation 2	SPEED drift near zero on all paraphrases; VG-ness unchanged (Section 11)

13.2 Implications for the Field

The two limitations we identify are not bugs — they are architectural consequences of SPEED’s design choices:

1. **Rank saturation** is inherent to any method that operates by projecting into the null-space of a finite-dimensional retain set. As the erased concepts become more entangled with the retain set, the null-space shrinks until it can no longer accommodate a non-trivial edit. This is a fundamental capacity constraint, not an implementation error.
2. **Lexical overfitting** is inherent to any method that operates on text embeddings of specific target tokens. Because the projection targets a specific direction in text-embedding space, it has no effect on alternative descriptions that map to nearby but distinct directions. Fine-tuning methods like ESD-x avoid this because they modify the model’s internal representation of the visual features, not just its response to specific text tokens.

13.3 Directions for Future Work

- **Rank saturation mitigation.** Methods that operate in a disentangled latent space (e.g., SAEmnesia, ICML 2026) may avoid the rank saturation problem by isolating concept representations into individual latent dimensions before erasing them.
- **Semantic-level erasure.** Methods that identify and erase the visual *features* of a concept rather than its text *tokens* would resist paraphrase evasion. ESD-x’s fine-tuning approach is a partial solution; a closed-form version that achieves semantic-level erasure while maintaining SPEED’s efficiency remains an open problem.
- **Inverting the rank saturation experiment.** Erasing 40 *pointillists* and predicting that Seurat becomes the casualty while Pissarro survives would provide the clean geometric confirmation of our directional analysis.

14. Conclusion

SPEED (Li et al., ICLR 2026) is a genuinely strong contribution to the concept erasure literature. Its training-free, closed-form editing achieves remarkable efficiency (seconds per edit vs. minutes for

fine-tuning methods), and its null-space guarantee provably preserves the retain set under normal operating conditions. Our sparse multi-concept erasure experiments confirm that the guarantee holds, and our LPIPS analysis confirms it operates at the perceptual level.

However, SPEED's architecture contains two concrete, reproducible limitations:

1. **Rank saturation collapse.** Under concentrated mass erasure of a semantically dense cluster (40 impressionists), the null-space shrinks until the most entangled retained concept (Pissarro) drifts to $2\times$ the control level. SPEED's own DPA refinement module actively accelerates this failure by consuming rank budget.
2. **Lexical overfitting.** SPEED's erasure is tied strictly to specific text tokens. Describing a target style without using the target name trivially bypasses the erasure, with zero image drift on paraphrased prompts. ESD-x, which fine-tunes cross-attention weights, suppresses paraphrased styles substantially more.

As long as erasure operates by projecting in cross-attention text-embedding space, the most entangled neighbors will be the first to leak under heavy load, and the most creative prompters will be the first to evade it. These are directions future disentanglement-based and semantic-level methods will need to address.

15. References

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All code, SLURM configurations, generated images, and raw analysis logs are available on the project repository branches: `main`, `all-experiments`, `experiment-3`, `experiment-3-rank-saturation`, and `experiment-4-paraphrase-evasion`.